

Midterm project

Project Plan for Capstone 1: System Analysis and Design

Topic: Online Study App (Loro Topik / Korean Learning App)

Team Name: 17

Team Members (4 members):

1. **Member 1** Jumayev Toxirjon.ID:202590182 (team leader).
2. **Member 2** Kazakov Arsen.ID: 202590187 (second leader)
3. **Member 3** Raxmatullayev Muhammadali (Diagram Designer)
4. **Member 4** Uqazbergenov Sunetulla.ID:202590241 (PPT Maker)

1. Project Plan

Project Objective

Develop an Online Study App (Loro Topik) that allows users to register, study Korean vocabulary and grammar, take TOPIK practice quizzes, track learning progress, and manage personal study schedules.

Project Scope

The system will cover:

- User registration and login
- Vocabulary and grammar study modules
- TOPIK practice tests and quizzes
- Learning progress tracking and statistics
- Bookmark / Favorites management
- Basic content management (Admin)

Target Users

- **Student / Learner:** Registers, logs in, studies vocabulary/grammar lessons, takes practice tests, and views progress statistics.
- **Administrator:** Manages study content, vocabulary databases, test questions, and user accounts.

2. Use Case Modeling

(Required: More than 3 Use Cases)

Planned Use Cases

1. Register Account
2. Login
3. Browse & Study Lessons (Vocabulary / Grammar)
4. Take TOPIK Practice Test
5. View Progress & Statistics
6. Bookmark Lessons / Words
7. Manage Study Content (Admin)
8. Manage User Accounts (Admin)

Diagram Explanation

The Use Case Diagram will show the main actors and their interactions with the Online Study App. The Student will interact with registration, study modules, practice tests, and progress tracking, while the Administrator will manage learning contents and user data.

Use Case Specifications

Each use case will have a short specification describing:

- Actor
- Precondition
- Main flow
- Postcondition

3. Business Process Model (BPM)

(Required: More than 2 BPMs)

- **BPM 1 – User Registration & Onboarding Process:** Shows the workflow from opening the app to entering user details, selecting target TOPIK level, creating the account, and receiving confirmation.
- **BPM 2 – Lesson Studying & Quiz Process:** Shows the workflow from selecting a lesson module, viewing study cards/grammar, taking a mini-quiz, saving scores, and updating learning progress.
- **BPM 3 – TOPIK Practice Test & Grading Process:** Shows how a student selects a mock test, answers questions, submits the test, and how the system calculates scores and generates performance reports.
- **BPM 4 – Content Management Process (Admin):** Shows how the Administrator logs in, creates or updates study materials (words, audio, questions), and publishes them to the app.

4. Chen ERD

(Required: More than 3 Chen ERDs)

- **Chen ERD 1 – User & Account:** Shows the relationship between User, Account, and User Profile, focusing on registration and personal details.
- **Chen ERD 2 – Study Module & Lesson Content:** Shows the relationship between Category, Lesson, Vocabulary, and Grammar, focusing on how learning materials are organized.
- **Chen ERD 3 – Test & Quiz Results:** Shows the relationship between User, Practice Test, Question, and User Score, focusing on quiz submission and progress evaluation.
- **Chen ERD 4 – User Bookmark & History:** Shows the relationship between User, Saved Word/Lesson, and Study History.

5. IE ERD

(Required: More than 3 IE ERDs)

- **IE ERD 1 – User and Account:** Shows the database relationship between User, Account, and Profile tables using primary and foreign keys.
- **IE ERD 2 – Category, Lesson, and Vocabulary:** Shows how course categories, individual lessons, vocabulary words, and grammar rules are connected in the database.
- **IE ERD 3 – User, Test, Question, and Score:** Shows how users take tests, how test items/questions are structured, and how score records are linked to a user.
- **IE ERD 4 – User, Bookmark, and Study Log:** Shows how user interaction logs and saved items are stored in relational tables.

6. Schedule

Stage	Possible Time	What Will Be Done	Reason
1. Project Plan	Week 1	Define objectives, scope, and target users	This should be done first because it establishes what the system will include and who will use it.
2. Use Case Modeling	Week 1-2	Create Use Case Diagram and Use Case Specifications	Use cases identify the system's main functions and user interactions before creating process and database models.
3. BPM	Week 2-3	Create the Business Process Models	BPMs should follow the use cases because they describe how the identified activities happen step by step.
4. Chen ERD	Week 3-4	Create more than 3 Chen ERDs	Chen ERDs can be developed after understanding the system processes because the processes help identify the required

Stage	Possible Time	What Will Be Done	Reason
			entities and relationships.
5. IE ERD	Week 4-5	Create more than 3 IE ERDs	IE ERDs can be created after the Chen ERDs because they provide a more structured representation of the same database relationships.
6. Review and Finalization	Week 5	Check all diagrams, specifications, relationships, and requirements	The final review ensures that all required deliverables are complete and consistent with each other before submission.

Overall Schedule: Approximately 5 weeks.

Reason for the Order:

The project starts with the Project Plan, then moves to Use Cases, Business Processes, Chen ERDs, and finally IE ERDs. Each stage provides information needed for the next stage, making the development process more organized and reducing inconsistencies between the diagrams.